

PoC Testing Report: automated corridor geometry, standards flags and vehicle passability from mobile LiDAR

Netrica.AI Private Limited with the GeoAI4Cities Laboratory, IISER Bhopal · Report version 1.0 · 4 October 2026 · SPOC Pyare Lal · netrica.ai@gmail.com · Live application: [this site](#) (static, no backend).

This report states what was tested, on which data, against which ground truth, and with what result, for every component that runs in the proof of concept. All figures come from the same pipeline and the same three Indian mobile-LiDAR surveys that the live application serves; the per-path numbers can be re-derived in the application's corridor-audit viewer, and the city-level shares are recomputed in the browser on the PoC home page.

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1 · Summary of results

0.10 m

Width mean absolute error vs hand-traced boundary, main spine (n = 3,982 stations); 0.19 m on branches (n = 243)

0.969

Carriageway boundary IoU vs 13 hand-traced polygons (main spine); 0.947 on branches

0.80 / 0.85

Junction completeness / correctness vs 215 hand-labelled junctions; type agreement 86%

67–80 mIoU

12-class segmentation after fine-tuning, per city (Bhopal 71.3, Kerala 67.6, Chandigarh 79.9); overall accuracy 92–96%

≤ 5 mIoU

Cost of deploying one merged encoder per city instead of city-specific fine-tuning, without sharing raw scans (naive transfer loses up to 38)

< 3 ms

Accessibility-gated route (A*) on a 1,364-node city graph, computed in the browser

COMPONENT	TEST	GROUND TRUTH / REFERENCE	RESULT	STATUS
Semantic segmentation	Held-out test paths, 12 classes, three cities, three PTV3 encoders	Manual point-wise labels	mIoU 66.7–79.9 fine-tuned; dominant classes 83–98% IoU; Wire 83–88%	Pass
Cross-city deployment	Merged encoder per city vs city fine-tune	Same test splits	Within 5 mIoU everywhere (Utonia: -0.6 / +1.8 / +0.2)	Pass
Carriageway boundary	13 traced polygons, 8 road-geometry types	Hand-digitised polygons on the MLS reference	IoU 0.969 main spine, 0.947 branches	Pass
Width per station	4,225 stations across three sites	Width from the traced polygons	MAE 0.10 m (spine), 0.19 m (branch); conservative bias	Pass
Junction detection	215 junctions, all 8 Bhopal drives + Kerala + Chandigarh paths	Hand-labelled points with types	Completeness 0.80, correctness 0.85, quality 0.70, type 0.86	Pass (over-detects on ring topologies)
Overhead clearance	Wire spans fitted per station stream	Fit residual gate	Quadratic per span, RMSE $\leq$ 0.25 m accepted; stations with no overhead evidence reported, never assumed clear	Pass
Deployed run on predicted labels	Bhopal drives 7 and 8, three encoders	Hand-labelled run of the same drives	Main-spine width error 0.58–1.00 m; branches 1.09–1.40 m → only the main spine is certified	Pass with scope rule
Routing engine	60 origin–destination pairs per city, A* vs Dijkstra vs SSSP	Exact shortest-path distances	Identical distances (A* within 0.5 m graph resolution); 0.25–2.5 ms per query	Pass
Street-view uplift (pilot)	76 annotated crops	Manual masks	Road 92.7–92.8% IoU, Wire 25.6–37.8%	Not certified · in development

2 • Test data and ground truth

Three mobile-laser-scanning (MLS) surveys of contrasting Indian urban forms, each manually labelled point by point and represented in a common 12-class taxonomy (ground, vegetation, building, road, wire, pole/tower, wire-structure connector, vehicle, board, fencing, human/animal, road-side object). Bhopal's labels follow a 62-category static taxonomy mapped hierarchically to the 12 classes. Train / validation / test splits are by complete survey paths, so test paths are spatially separate from training paths.

SITE	URBAN FORM	PLATFORM & SCANNER	YEAR	DENSITY (PTS/M ²)	COLOUR	ROUTABLE MAIN SPINE	STATIONS IN THE APP
Bhopal (BM)	Dense historic core, bazaar lanes, low wiring	Handheld GeoSLAM ZEB Horizon (Velodyne VLP-16 + 360° camera), DGPS georeferenced; 8 drives	Mar 2023	2,829	RGB on 35–40% of points	10.6 km	2,427
Kerala (KM)	Semi-urban, vegetated, narrow village lanes	Vehicle-mounted Riegl VQ-450	2020	1,787	intensity only	8.1 km	1,727
Chandigarh (CH)	Planned grid, wide avenues, mature tree canopy	Vehicle-mounted StreetMapper, dual Riegl VQ-450 + GNSS/IMU (Geokno, for Chandigarh Administration)	2020	2,224	intensity only	17.2 km	4,054

Densities under a common 5 m tile / 25 cm voxel protocol (mean point spacing 4.9 / 4.9 / 4.0 cm). "Stations in the app" = measured 5–10 m carriageway stations in the layers served by the live application (main spine and branches).

Ground truth used for validation

- **Semantic labels:** manual point-wise annotation of every survey (Bhopal in 62 categories; Kerala's labels ship with the dataset; Chandigarh follows the Kerala annotation and quality-control protocol).
- **Carriageway polygons:** 13 hand-traced polygons over the MLS reference across the three sites, covering eight road-geometry classes: simple corridor, organic loop, dense street network, four-arm roundabout, junction, straight segment, 90° corner, divided dual carriageway.
- **Junctions:** 215 hand-labelled junction points with type (142 three-way T/Y, 56 four-way, plus skew and ring cases) spanning all eight Bhopal drives and the traced Kerala and Chandigarh paths.
- **Per-station checks:** hand-measured ground truth on stratified stations is used to check the thresholds behind the standards tiers.

3 • Semantic segmentation accuracy

Three Point Transformer V3 foundation encoders (Sonata, Concerto, Utonia) were evaluated under three protocols: linear probe (frozen encoder, linear head), decoder probe (frozen encoder, trained decoder) and full fine-tuning. Input is a 9-channel representation (coordinates, colour where present, intensity, normals), native resolution, test-time augmentation. The table reports the fine-tuning protocol used in deployment.

MODEL (FINE-TUNED, 12 CLASSES)	PARAMS (M)	BHOPAL OA	BHOPAL MIOU	CHANDIGARH OA	CHANDIGARH MIOU	KERALA OA	KERALA MIOU
Sonata	124.8	92.12	68.72	95.89	79.60	93.87	67.52
Concerto	224.2	93.42	71.06	95.80	79.92	92.40	67.58
Utonia	162.2	92.69	71.28	96.36	79.76	92.82	66.66

OA = overall accuracy, mIoU = mean intersection over union over the 12 classes, both in %. Linear and decoder probes score 55.7–77.1 mIoU and are not deployed.

Per-class behaviour. The classes the corridor measurement depends on are segmented reliably everywhere under fine-tuning: Vegetation 96–98% IoU, Building 83–97%, Road 86–93%, and Wire 83–88% despite being a thin structure. The hardest classes are the rare, city-dependent ones: Wire–Structure Connector reaches 79–85% in Bhopal (dense overhead wiring) but is absent from Chandigarh and sparse in Kerala (24–49%); Road–Side Object is near zero in Kerala. These near-empty classes pull Kerala's mean below Bhopal's although its scenes are visually simpler.

4 • Cross-city transfer and model merging

A model fine-tuned on one city transfers poorly to another: the best single-source model into Bhopal reaches 43.4 mIoU against the in-city 71.3 (a loss of up to 38 points across pairs). The deployment mechanism is therefore **multi-city model merging**: fine-tuned encoders are combined in weight space into one encoder per target city, and each city's own segmentation head is re-attached. Raw scans are never pooled, so the method works where data cannot be shared between owners.

ENCODER	BHOPAL MERGED	BHOPAL FT	Δ	KERALA MERGED	KERALA FT	Δ	CHANDIGARH MERGED	CHANDIGARH FT	Δ
Sonata	65.03	68.72	−3.69	68.55	67.37	+1.18	79.46	79.50	−0.04
Concerto	68.74	68.10	+0.64	66.63	63.57	+3.06	78.61	78.88	−0.27
Utonia (deployed)	70.66	71.28	−0.62	68.14	66.35	+1.79	79.84	79.66	+0.18
Utonia, one universal encoder	61.21	71.28	−10.07	68.04	66.35	+1.69	78.61	79.66	−1.05
Utonia, pooled-data joint (reference)	69.80	71.28	−1.48	75.73	66.35	+9.38	80.62	79.66	+0.96

mIoU (%) on each city's held-out test split. FT = within-city fine-tuning. Merged = per-city merged encoder. The universal row ships one encoder unchanged to all three cities; the joint row pools raw data (strong for the smallest city, but requires centralised scans and a full retrain per added city).

What this means for a Rodic corridor. A new city gets a merged encoder built from the existing fine-tunes plus a small in-city fine-tune when labels exist; no partner data has to leave its premises to benefit from the three-city training base. The expected cost of merging is bounded here at about 5 mIoU against a full in-city fine-tune.

5 • Geometric validation

The width chain (carriageway boundary → centreline → junctions → width per station) was validated against hand-digitised ground truth on three fronts, with main spine and branches reported separately because the system certifies only the main spine (section 6).

PATH	BOUNDARY IOU	WIDTH MAE (M)	GT JUNCTIONS	TP	FP	MISSED	TYPE ACC.
Bhopal 1 (dense organic network)	0.965	0.18	26	17	8	9	0.76
Bhopal 2	0.971	0.12	32	29	1	3	0.97
Bhopal 7	0.975	0.13	20	15	3	5	0.87
Bhopal 8	0.919	0.17	17	17	5	0	0.88
B3S1 · B3S2 · B3S3 (Bhopal short corridors)	0.978 · 0.955 · 0.961	0.14 · 0.18 · 0.14	4 · 3 · 9	3 · 1 · 4	0 · 0 · 0	1 · 2 · 5	1.00 · 0.00 · 0.50
Kerala 1 · 2 · 3 · 4	0.980 · 0.986 · 0.986 · 0.979	0.07 · 0.05 · 0.05 · 0.07	0 · 0 · 1 · 1	— · — · 1 · 1	0	0	1.00
Chandigarh 1 (divided dual carriageway) · 2	0.967 · 0.973	0.20 · 0.13	12 · 0	12 · —	5 · 1	0 · —	0.92 · —
Bhopal 3 · 4 · 5 · 6 (junction GT only)	—	—	23 · 22 · 34 · 11	16 · 15 · 30 · 10	0 · 5 · 2 · 0	7 · 7 · 4 · 1	0.75 · 0.80 · 0.93 · 0.80
Main spine	0.969	0.10 (n = 3,982)					
Branches	0.947	0.19 (n = 243)					
All junctions			215	171	30	44	0.86

Junction matching is one-to-one within 8 m. Completeness 0.80, correctness 0.85, quality 0.70. An earlier evaluation restricted to the thirteen traced paths (125 junctions) gave 0.80 / 0.81 / 0.68 and type 0.87, so the detector's accuracy is stable across validation-set size.

- **Width error is conservative.** The recovered carriageway is marginally narrower than truth, so certification errors understate rather than overstate access. The 0.10 m spine error sits well inside the ≥ 0.5 m gaps between vehicle-

class thresholds (3.0 / 3.5 / 5.5 / 7.0 m).

- **Denosing was tested and rejected.** Three outlier filters (statistical, density/off-surface, detachment guard) scored against the traced polygons on nine paths improved no path by more than a millimetre and worsened width error on the five denser paths by factors of 1.05–2.7; clouds are therefore measured raw, at full resolution.
- **Junction detector choice.** On a 79-junction subset the training-free geometric detector (orientation-field defects) scores quality 0.671 against 0.548 for a detector on learned 3D features, with correctness 0.905 against 0.785; the geometric detector is the one deployed.
- **Residual junction error** is dominated by over-detection on ring topologies (four-arm roundabout, organic old-city loop). Both failure modes fail safe: a spurious junction disc masks only a short stretch, and a missed junction can only widen, never narrow, the measured carriageway.

6 • Behaviour on model-predicted labels (deployed condition)

Everything above uses hand labels. A new corridor will be measured on *predicted* labels, so the full chain was rerun on the two held-out Bhopal drives (7 and 8) with three encoders (12-class fine-tune, 21-class fine-tune, merged) and compared with the hand-labelled run of the same drives. The per-path shares below are the ones shown in the application's corridor-audit *Deployment* panel.

DRIVE • LABELS	LENGTH (M)	MIN WIDTH (M)	MEDIAN WIDTH (M)	AMBULANCE CAN USE	FIRE TRUCK CAN USE	AERIAL CAN USE
Bhopal 7 • hand labels	1,314	1.97	4.68	78.7%	53.2%	41.2%
Bhopal 7 • predicted, 12-class encoder	1,357	0.80	4.04	81.8%	49.5%	37.6%
Bhopal 7 • predicted, 21-class encoder	1,360	1.19	4.38	80.9%	54.1%	40.7%
Bhopal 7 • predicted, merged encoder	1,376	1.60	4.03	79.5%	46.8%	33.6%
Bhopal 8 • hand labels	930	1.07	4.05	67.1%	29.6%	20.0%
Bhopal 8 • predicted, 12-class encoder	936	0.98	3.65	63.3%	28.9%	20.0%
Bhopal 8 • predicted, 21-class encoder	935	1.10	3.99	67.5%	29.0%	20.0%
Bhopal 8 • predicted, merged encoder	995	1.09	3.37	56.4%	26.7%	20.4%

Shares are of measured station length passable to each vehicle on that drive (per-vehicle width and clearance thresholds), as computed by the pipeline and listed in the application's validation manifest. Hand-label rows are the reference.

Width error under predicted labels. Pooled over the two drives, main-spine width error is 0.78, 0.58 and 1.00 m for the 12-class, 21-class and merged encoders, and branch error 1.18, 1.09 and 1.40 m (over 3 m on the branch-sparse drive 7). Branch error is comparable to the gaps between vehicle-class thresholds. **Policy adopted:** only the main spine is certified; side-street stubs seen one-sidedly are reported as width-uncertain (WU / AU) and excluded from accessible-length denominators and from routing. This rule is applied identically in the live application.

Choosing an encoder for accessibility, not just mIoU. Across twelve encoder variants the one with the highest mIoU (71.3%) ranked only ninth on accessibility fidelity (mean absolute error of the five accessibility-class shares against the hand-label run: 3.5 pp), while the encoders that track the hand-label accessibility map best are not the mIoU leaders. The deployed configuration is selected on accessibility fidelity.

7 · Width, clearance and accessibility results

CITY	AVG. WIDTH (M)	STATIONS < 3.0 M WIDE	OVERHEAD OBJECT PRESENT	CLEARANCE < 4.5 M	CLEARANCE < 3.5 M
Bhopal	4.9	7.5%	88.9%	56.3%	24.6%
Kerala	4.7	9.1%	69.4%	21.0%	1.3%
Chandigarh	8.2	4.7%	62.1%	37.7%	12.1%

All measured carriageway stations, main spine and branches, hand-labelled clouds. Clearance = lowest of the wire, structure and vegetation streams over the drivable band.

CITY (ROUTABLE MAIN SPINE)	AMBULANCE (3.0 / 3.5 M)	FIRE TRUCK (3.5 / 4.5 M)	AERIAL APPLIANCE (3.5 / 5.0 M)	BINDING OBJECT AT OBSTRUCTED STATIONS
Bhopal · 10.6 km	72.9%	40.1%	27.9%	Wire at 79% (low bazaar wiring; shop shades, boards)
Kerala · 8.1 km	90.9%	64.4%	54.2%	Mixed: utility wires and tree canopy
Chandigarh · 17.2 km	85.0%	58.3%	53.4%	High vegetation at 84% (avenue-tree canopy over the drive lane)

Share of certifiable road length accessible to each vehicle class (width / clearance thresholds in the column heads).

Which axis blocks each vehicle. For the large apparatus the overhead axis blocks more road than the width axis at every site: among stations failing on one axis only, overhead exceeds width for the fire truck (64.9 vs 5.2% of blocked stations in Bhopal; 45.4 vs 40.9 in Kerala; 87.2 vs 7.1 in Chandigarh) and by a wider margin for the aerial appliance (70.9 vs 2.8; 57.4 vs 25.0; 88.6 vs 5.7). The ambulance is the exception in Kerala, where narrow village lanes (width) dominate. A wide planned carriageway therefore does not by itself guarantee vertical clearance: Chandigarh's aerial-appliance share (53.4%) equals Kerala's (54.2%) despite carriageways nearly twice as wide.

Live recomputation. The PoC home page recomputes length-weighted A-class shares, overhead presence and the binding object from the served station layers (all measured stations, including branches, with WU reported separately). Those shares therefore differ slightly in scope from the main-spine table above; both are correct for their scope.

8 · Accessibility-gated routing

Emergency response was simulated over facility → incident pairs on the aligned OSM mesh carrying the per-edge codes: origins are the OSM hospitals and fire stations nearest each surveyed area (23 / 11 / 44), destinations 30

incident points per city on surveyed streets. Each edge is gated on the minimum width and clearance along it. Unsurveyed road is treated as passable, so detours are lower bounds, and the share of each gated route on unsurveyed road is reported.

CITY	VEHICLE	INCIDENTS UNREACHABLE	NO ROUTE (OF SERVICEABLE PAIRS)	REROUTED	MEDIAN DETOUR	MEDIAN DELAY (MIN)	ROUTE ON UNSURVEYED ROAD
Bhopal	Ambulance	53.3%	11.0%	57.1%	27.7%	0.72	66.5%
	Fire truck	66.7%	70.6%	14.7%	8.9%	0.29	100%
	Aerial appliance	70.0%	75.4%	11.5%	8.5%	0.32	100%
Kerala	Ambulance	23.3%	36.1%	46.2%	26.5%	3.11	86.9%
	Fire truck	43.3%	64.0%	21.1%	12.3%	1.95	100%
	Aerial appliance	43.3%	64.0%	21.1%	12.3%	1.95	100%
Chandigarh	Ambulance	16.7%	65.5%	18.2%	23.6%	2.83	100%
	Fire truck	23.3%	73.3%	11.3%	24.5%	1.19	100%
	Aerial appliance	23.3%	73.3%	11.3%	24.5%	1.19	100%

Survey coverage of the routing mesh: Bhopal 69.3% (one traversable corridor, 15.2 km), Kerala 29.6% (2 components, 16.4 km), Chandigarh 19.4% (16 components, 28.3 km). The worst Bhopal ambulance dispatch runs 589 m unrestricted but 2.49 km once gated (+323%, about 3.5 min).

CITY GRAPH	NODES	EDGES	A* (MS)	A* NODES EXPANDED	DIJKSTRA (MS)	DIJKSTRA NODES EXPANDED
Bhopal	252	718	0.25	43	0.39	118
Kerala	239	550	0.27	54	0.35	123
Chandigarh	1,364	3,842	1.49	252	2.47	691

60 origin–destination pairs per city. SSSP distances are identical to Dijkstra's and A* agrees within the 0.5 m graph resolution. The live application runs the same Dijkstra / A* in the browser (no server round trip) and reports nodes expanded per query in its route card.

9 • Runtime, footprint and the delivered application

- **Delivery:** a static web application (HTML, TypeScript compiled to JavaScript, CesiumJS 3D Tiles, three.js point clouds, GeoJSON layers). No backend, no database, no cookies; hostable on any static host or object store, including on-premise.
- **Payload:** under 1 GB for the three-city PoC including classified point clouds, 3D Tiles, network layers, routing graphs and validation geometry; individual files under 25 MB; heavy assets load lazily as a section scrolls into view.

- **Browser requirements:** a WebGL 2 browser (Chrome, Edge, Firefox, Safari 16+); the audit viewer is designed for laptop or larger screens.
- **Batch pipeline:** Python / PyTorch; segmentation on one NVIDIA A100 / L40S-class GPU; geometry, standards and routing steps CPU-only. Merged inference costs the same as a single model.
- **Throughput on a Rodic corridor** will be measured during the PoC run and reported with the corridor results; the application's acceptance target is end-to-end processing under one hour per 10 km on a single GPU.

10 • Street-view imagery pilot (not certified)

A pilot tested whether street-view crops can be lifted to metric 3D and segmented with the same encoders, as a cheap signal for *change* between LiDAR surveys. On 76 annotated crops the road surface is recovered well (92.7–92.8% IoU) but Wire collapses to 25.6–37.8% against 83–88% on LiDAR. Because wire sets the clearance that gates the largest vehicles, this path cannot certify geometry. **It is excluded from the PoC's certified outputs** and is positioned only as a trigger for re-survey.

11 • Limitations and failure modes

- **Scope of certification.** Only the routable main spine is certified; side-street stubs with one-sided evidence are reported as uncertain and never routed.
- **Survey coverage bounds routing claims.** Where a survey covers a small fraction of the mesh (Chandigarh 19%), gated routes run largely on unsurveyed road; detour figures are lower bounds.
- **Thin-structure accuracy.** Wire is the accuracy-critical class at 83–88% IoU; a dedicated power-line extraction stream is the natural upgrade.
- **Transient objects.** Parked vehicles narrow the measured carriageway at capture time; the ground-cause attribution names them and repeat drives reconcile them, but a single drive reports the street as it was that day.
- **Ring topologies** (roundabouts, old-city loops) produce extra junction detections; accepted because the failure is safe by construction.
- **Not implemented:** road gradient / slope; instance-level asset register export in the browser (scheduled for the Rodic corridor run as a GeoPackage deliverable).

12 • Reproducibility and traceability

- **Per-path numbers** (boundary, width per station, clearance per stream, junctions, verdicts) can be inspected in the application's corridor-audit viewer; ground-truth polygons and junction labels overlay on the validation paths.
- **City-level shares** are recomputed in the browser from the served station layers (`data/network/<city>_categorization.geojson`, EPSG:32643) on the PoC home page.
- **Routing** is executed live over the served graphs (`data/network/<city>_graph.json`); the route card reports lengths, detour and nodes expanded for each query.
- **Peer review.** The methodology and the tables in sections 3–8 are from a manuscript by the team currently under review at a leading photogrammetry and remote-sensing journal; numbers here follow the manuscript revision of 29 September 2026.
- **Data availability for evaluators.** The three surveys are research datasets of IISER Bhopal and partners (Bhopal: own capture; Kerala: published dataset; Chandigarh: Chandigarh Administration survey processed under agreement); the served layers are derived products and carry no personal data. Access to source clouds can be arranged for the jury under the owners' terms.

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